

Targeted Analytical Strategy for Arabidopsis Radiation Response

Cross-Study Integration of GO Enrichment from 11 Comparisons Spanning 10 cGy to 100 Gy

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Context: This document analyzes GO enrichment results from 11 comparisons across four NASA GeneLab studies (OSD-498, OSD-510, OSD-508, OSD-658, OSD-782), covering gamma radiation at 10 cGy to 100 Gy and simulated GCR at 40-80 cGy, including the *sog1_1* vs. WT genotype interaction. It identifies cross-study patterns, SOG1-dependent gene sets, dose-response trends, and proposes targeted analytical approaches to advance the work toward a publication-quality plant radiation biology study.

1. Executive Summary

The complete GO enrichment dataset comprises **11 comparisons** across four GeneLab studies, yielding a total of **1,377 enriched GO terms** backed by **5,248 unique genes**. The data spans five orders of magnitude in radiation dose (10 cGy to 100 Gy), two radiation qualities (gamma and simulated GCR), and includes a critical genotype interaction (*sog1_1* vs. WT). This represents one of the most comprehensive plant radiation transcriptomic datasets assembled for cross-study analysis.

Three major findings emerge from the cross-study analysis:

Finding 1: SOG1 dependency is directly demonstrated. The 5b interaction comparison (OSD-508/510) identifies **46 radiation-responsive genes whose induction is lost in the *sog1_1* mutant**, including 11 known SOG1 direct targets (RAD51, BRCA1, PARP1, PARP2, TSO2, XRI1, RAD17, POLK, SYN2, TIL1, TK1A). The interaction-down pathways are dominated by DNA repair, DNA damage response, double-strand break repair, and response to ionizing radiation — confirming that SOG1 is the master regulator of the DDR under actual gamma exposure, not just zeocin treatment [14].

Finding 2: The radiation response is bipartite and dose-dependent. Upregulated pathways consistently include hypoxia/oxygen response, defense/immune signaling, and programmed cell death — but the specific pathways differ by dose and radiation quality. At low gamma doses (10-100 cGy, OSD-782), hypoxia and defense dominate. At high gamma dose (100 Gy, OSD-498/510), defense and immune responses are strongest. For GCR (OSD-658), DNA repair is the exclusive upregulated pathway with extremely high fold enrichment (26-48x). Downregulated pathways are more conserved: ribosome biogenesis, rRNA processing, cell cycle, and plastid organization appear across gamma doses.

Finding 3: Gamma and GCR produce fundamentally different transcriptomic responses. Gene-level overlap between gamma and GCR studies is essentially zero (Jaccard < 0.005). GCR exposure (OSD-658) activates a narrow, DDR-focused response (2-7 genes, but 26-48x fold enrichment), while gamma exposure activates broad defense/hypoxia/cell-death programs (798-1,578 genes). This radiation-quality difference is one of the most striking findings and has direct implications for space radiation risk assessment.

2. Cross-Study Analysis of GO Enrichment Results

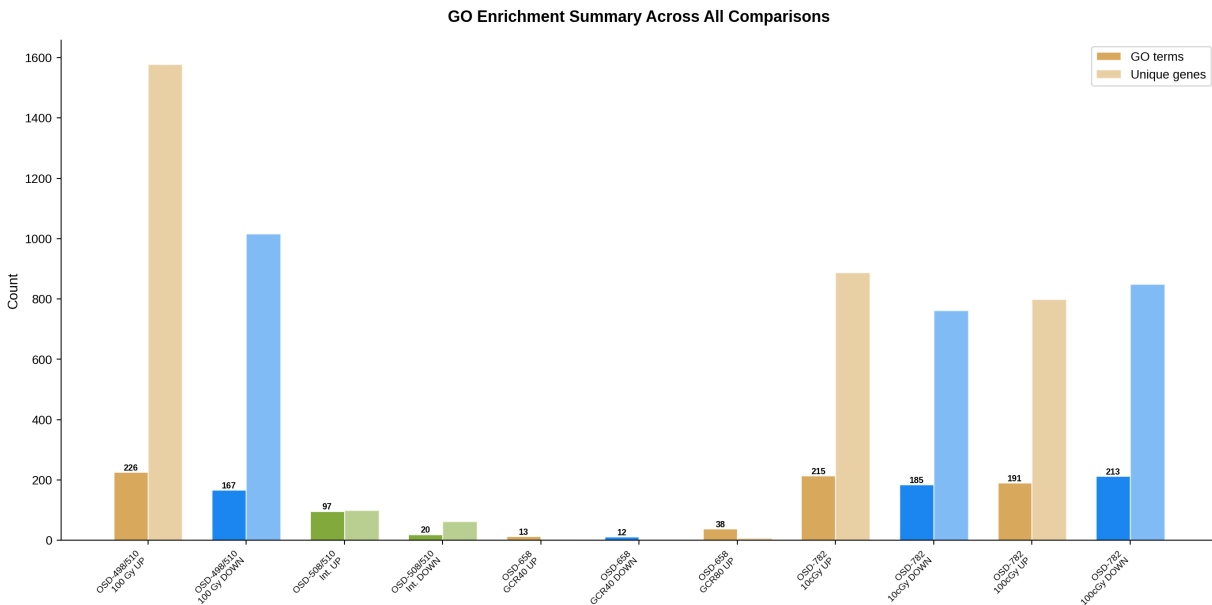


Figure 1: GO enrichment summary across all 11 comparisons. Gold bars = upregulated, blue = downregulated, green = genotype interaction. GO term counts (left bars) and unique gene counts (right bars) per comparison.

2.1 Dataset Overview

Study	Com p.	Radiation	Dose	Stage	Genotype	GO terms	Genes (UP /DOWN)	Role
OSD-498/510	5a	Gamma (Co60)	100 Gy	6-day seedlings	Col-0 WT + sog1_1	226 UP / 167 DOWN	1,578 / 1,016 genes	Radiation effect (main)
OSD-508/510	5b	Gamma (Co60)	100 Gy	6-day seedlings	sog1_1 x WT	97 UP / 20 DOWN	100 / 63 genes	SOG1 genotype interaction
OSD-658	5d	Simulated GCR	40 cGy	10-day seedlings	Col-0	13 UP / 12 DOWN	3 / 2 genes	Low-dose GCR
OSD-658	5d	Simulated GCR	80 cGy	10-day seedlings	Col-0	38 UP	7 genes	GCR dose-response
OSD-782	5c	Gamma (Cs137)	10 cGy	4-week rosettes	Col-0	215 UP / 185 DOWN	888 / 761 genes	Low-dose gamma
OSD-782	5c	Gamma (Cs137)	100 cGy	4-week rosettes	Col-0	191 UP / 213 DOWN	798 / 849 genes	Dose-response gamma

Table 1: Dataset overview. 11 comparisons from 4 GeneLab studies spanning 10 cGy to 100 Gy.

2.2 Pathway Conservation Across Studies

The heatmap below shows the presence and significance of 20 key pathways across all 11 comparisons. Several patterns are immediately visible: (1) hypoxia and defense pathways are upregulated across gamma doses but absent in GCR; (2) DNA repair is upregulated in GCR and high-dose gamma but appears in the downregulated set for low-dose gamma (OSD-782); (3) ribosome biogenesis, rRNA processing, and cell cycle are consistently downregulated across gamma doses; (4) the sog1_1 interaction specifically captures DNA repair and DDR as SOG1-dependent, and cell cycle as SOG1-independent.

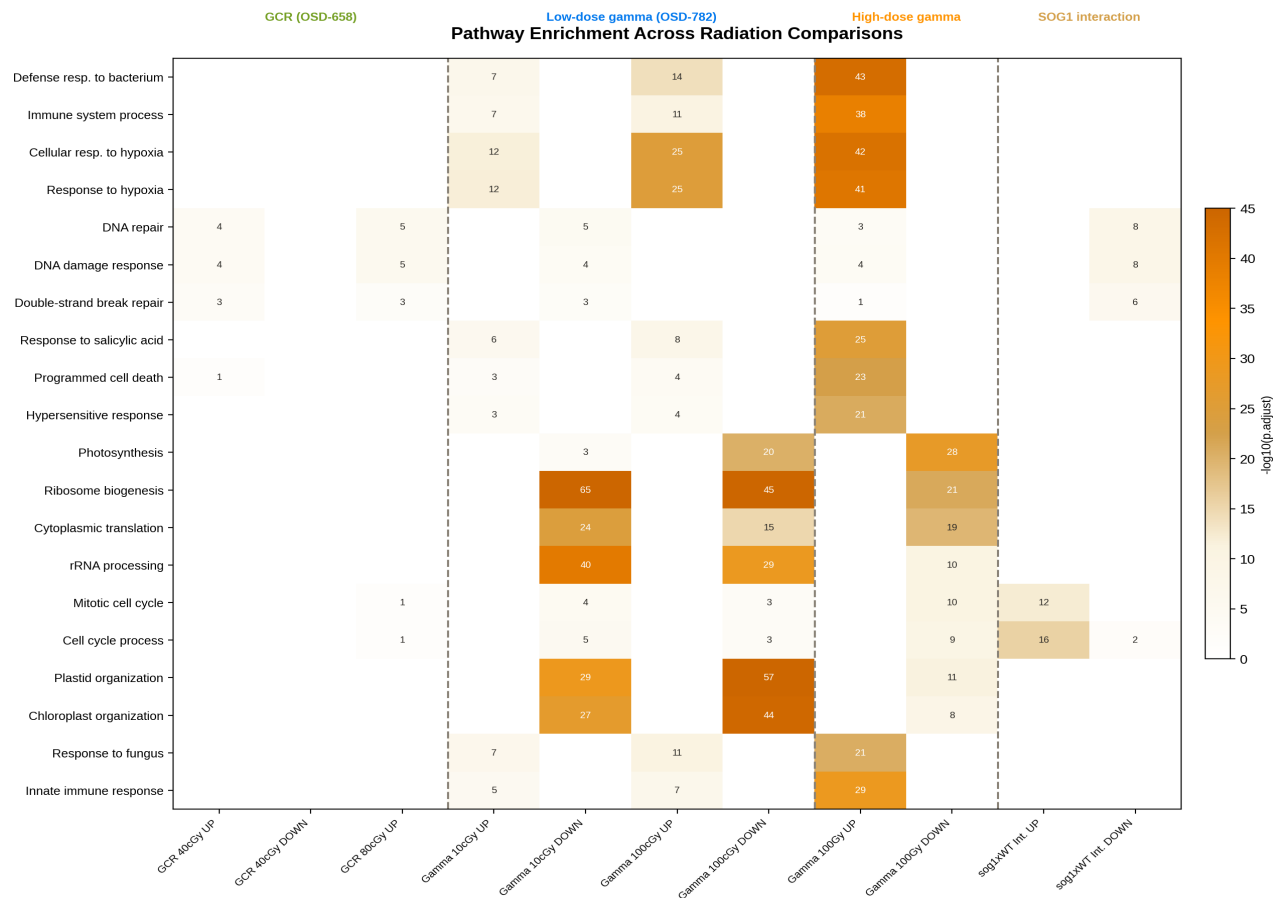


Figure 2: Pathway enrichment heatmap. Cell values are $-\log_{10}(p.adjust)$. White = absent. Dashed lines separate GCR (left), low-dose gamma (center-left), high-dose gamma (center-right), and SOG1 interaction (right) comparisons.

2.3 Conserved Pathways Across Radiation Studies

Only **2 GO terms** are upregulated in **4+ of 5 radiation comparisons**: programmed cell death (GO:0012501) and cell death (GO:0008219). This low overlap reflects the dose- and quality-dependent nature of the upregulated response. In contrast, **59 GO terms** are downregulated in **3+ of 4 comparisons**, indicating that the downregulated response is more conserved. The most consistently downregulated pathways are ribosome biogenesis, rRNA processing, cell cycle (G1/S transition, mitotic cell cycle), plastid organization, and chromosome organization.

Pathway	GO ID	Conservation	Present in	Biological interpretation
Programmed cell death	GO:0012501	4/5 UP	100Gy, 10cGy, 100cGy, GCR40	Core stress death response
Cell death	GO:0008219	4/5 UP	100Gy, 10cGy, 100cGy, GCR40	Core stress death response

Pathway	GO ID	Conservation	Present in	Biological interpretation
Ribosome biogenesis	GO:0042254	3/4 DOWN	100Gy, 10cGy, 100cGy	Translation suppression
rRNA processing	GO:0006364	3/4 DOWN	100Gy, 10cGy, 100cGy	Ribosome production halt
Cell cycle process	GO:0022402	3/4 DOWN	100Gy, 10cGy, 100cGy	Proliferation arrest
G1/S transition	GO:0000082	3/4 DOWN	100Gy, 10cGy, 100cGy	Cell cycle checkpoint
Plastid organization	GO:0009657	3/4 DOWN	100Gy, 10cGy, 100cGy	Chloroplast remodeling
Chromosome organization	GO:0051276	3/4 DOWN	100Gy, 10cGy, 100cGy	Genome stability

Table 2: Conserved pathways across radiation studies. Upregulated pathways show low conservation (max 4/5); downregulated pathways are more conserved (3/4).

2.4 SOG1 Regulatory Network: Direct Evidence from 5b Interaction

The 5b genotype interaction comparison (OSD-508/510) provides direct experimental evidence for SOG1 dependency. The interaction-DOWN set (genes whose radiation response is reduced in *sog1_1*) is dominated by DNA repair, DNA damage response, double-strand break repair, response to ionizing radiation, and homologous recombination — all canonical SOG1-regulated processes [14, 12]. The interaction-UP set (genes whose radiation response is retained or enhanced in *sog1_1*) is dominated by cell cycle, mitotic, and meiotic processes, suggesting these are SOG1-independent.

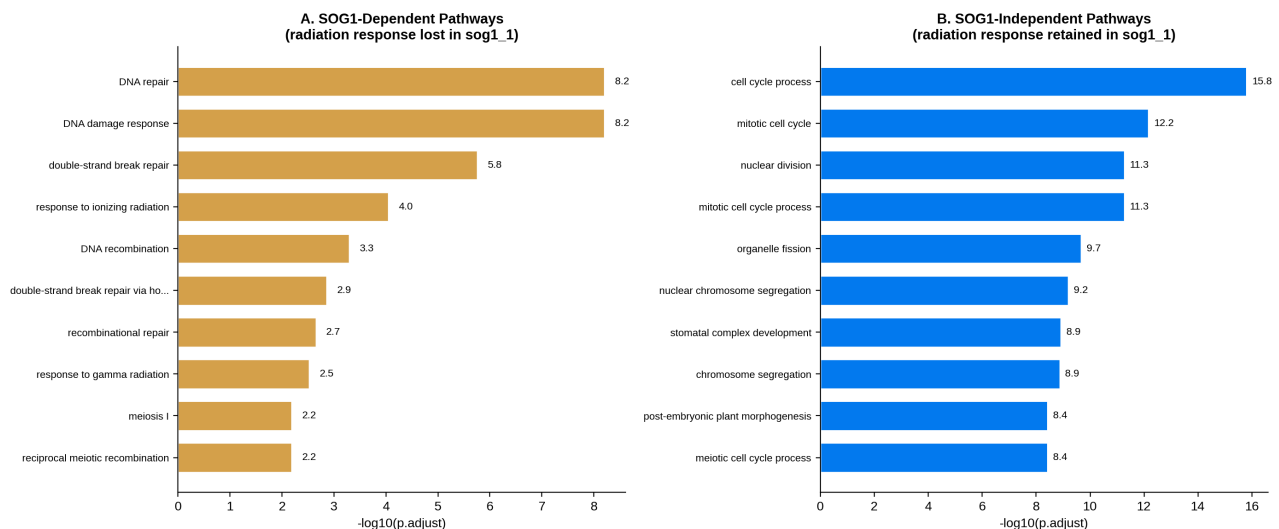


Figure 3: SOG1 genotype interaction (5b). Panel A: pathways whose radiation response is lost in *sog1_1* (SOG1-dependent). Panel B: pathways retained in *sog1_1* (SOG1-independent).

Cross-referencing the 5a upregulated genes with the 5b interaction-down genes identifies **46 SOG1-dependent radiation-responsive genes** — genes that are induced by radiation in WT but lose this induction in *sog1_1*. Critically, **11 of these are known SOG1 direct targets** from Ogita et al. (2018) [14]: RAD51, BRCA1, PARP1, PARP2, TSO2, XRI1, RAD17, POLK, SYN2, TIL1, and TK1A. The remaining 35 genes are novel SOG1-dependent candidates for follow-up validation.

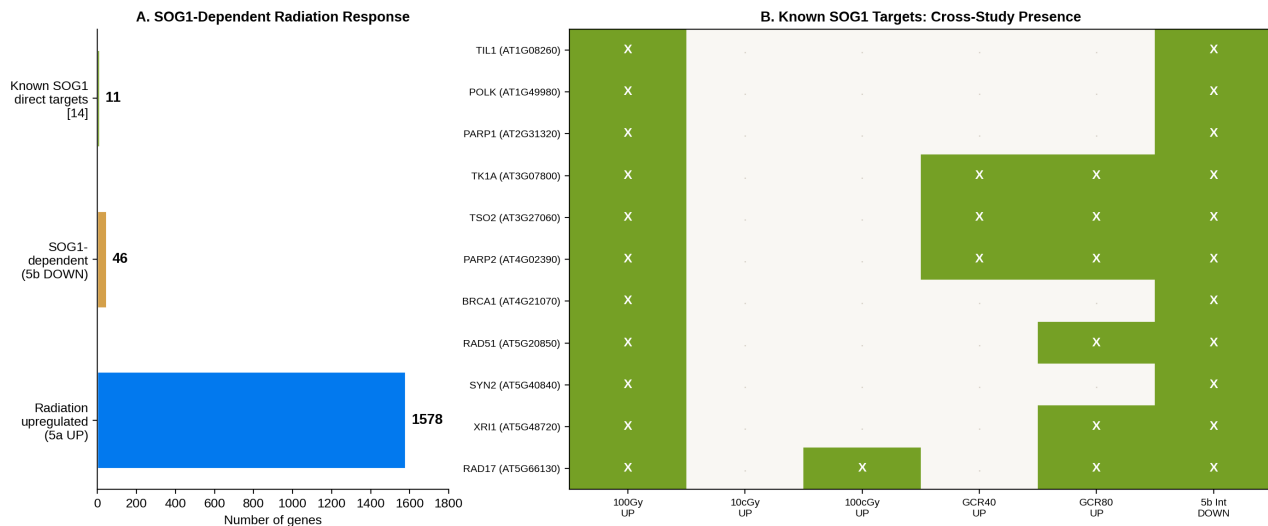


Figure 4: SOG1-dependent radiation response. Panel A: gene set sizes showing the convergence of radiation upregulation, SOG1 dependency, and known SOG1 targets. Panel B: cross-study presence of the 11 known SOG1 direct targets found in the data.

2.5 Known SOG1 Direct Targets Identified in the Data

Gene	AGI	Function	Present in comparisons
RAD51	AT5G20850	HR repair	100Gy UP, 5b DOWN, GCR80 UP
BRCA1	AT4G21070	HR repair	100Gy UP, 5b DOWN
PARP1	AT2G31320	DDR signaling	100Gy UP, 5b DOWN
PARP2	AT4G02390	DDR signaling	100Gy UP, 5b DOWN, GCR40/80 UP
TSO2	AT3G27060	Nucleotide synthesis	100Gy UP, 5b DOWN, GCR40/80 UP
XRI1	AT5G48720	DSB repair	100Gy UP, 5b DOWN, GCR80 UP
RAD17	AT5G66130	Checkpoint	100Gy UP, 5b DOWN, GCR80 UP, 100cGy DOWN
POLK	AT1G49980	Translesion synthesis	100Gy UP, 5b DOWN, 10cGy DOWN
SYN2	AT5G40840	Meiotic HR	100Gy UP, 5b DOWN
TIL1	AT1G08260	DNA replication	100Gy UP, 5b DOWN, 10cGy DOWN
TK1A	AT3G07800	Nucleotide synthesis	100Gy UP, 5b DOWN, GCR40/80 UP

Table 3: 11 known SOG1 direct targets [14] found among the 46 SOG1-dependent radiation-responsive genes. These genes are upregulated by radiation in WT and lose induction in sog1_1.

2.6 Dose-Response Patterns

The dose-response analysis reveals three distinct response regimes:

Low-dose gamma (10-100 cGy, OSD-782): Broad transcriptomic response (191-215 GO terms, 761-888 genes). Upregulated: hypoxia, defense/immune, protein phosphorylation. Downregulated: ribosome biogenesis (strongest signal, $p.adjust = 6.2e-66$ at 10 cGy), plastid organization, cell cycle. The 10 and 100 cGy responses share 148 upregulated and 124 downregulated GO terms (Jaccard 0.39/0.43), indicating a conserved low-dose gamma signature.

High-dose gamma (100 Gy, OSD-498/510): Even broader response (226 GO terms, 1,578 genes upregulated). Defense response to bacterium is the top pathway (193 genes, FE=2.8, $p=7.6e-44$). Hypoxia and immune signaling are strongly enriched. Gene overlap with low-dose gamma is moderate (Jaccard 0.16-0.18), sharing 341-360 genes — a core gamma response that scales with dose.

Simulated GCR (40-80 cGy, OSD-658): Narrow, DDR-focused response (13-38 GO terms, 2-7 genes). DNA repair is the exclusive upregulated pathway with extreme fold enrichment (26-48x). Glucosinolate biosynthesis is downregulated at 40 cGy (FE=323x, $p=5.2e-05$), confirming the Dixit et al. (2023) finding [2]. Gene overlap with gamma studies is essentially zero (Jaccard < 0.005), demonstrating that GCR and gamma produce fundamentally different transcriptomic responses despite overlapping dose ranges.

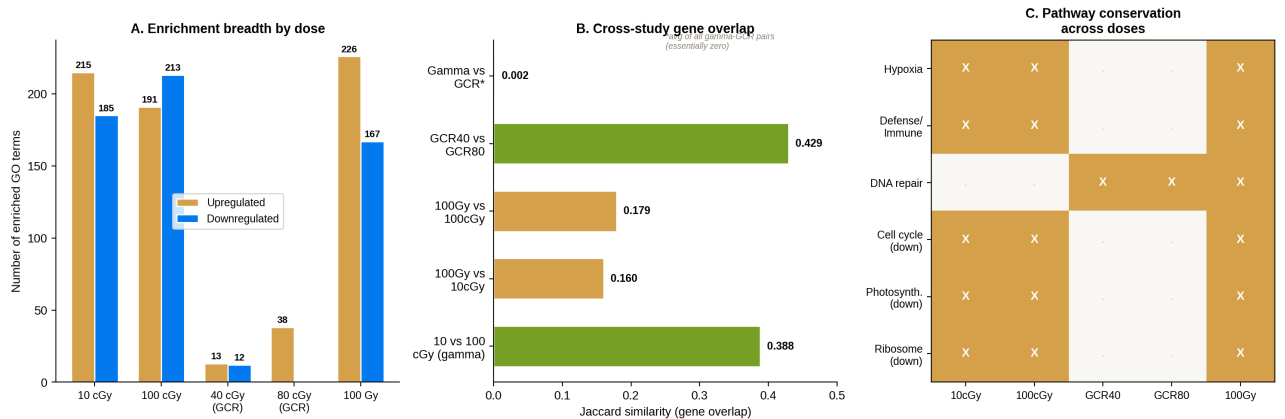


Figure 5: Dose-response summary. Panel A: GO term counts by dose and direction. Panel B: gene-level Jaccard similarity between study pairs. Panel C: pathway conservation matrix across dose levels.

3. Literature Synthesis: Grounding the Findings

The cross-study findings are grounded in a focused review of 18 key papers. The most relevant connections between the data and the literature are:

3.1 SOG1 as Master Regulator

Ogita et al. (2018) identified 146 direct SOG1 target genes using ChIP-seq and microarray in zeocin-treated seedlings [14]. The user's 5b interaction data now confirms SOG1 dependency under actual gamma radiation, with 11 of those 146 targets directly validated. Bourbousse et al. (2018) showed that SOG1 activates upregulated genes while MYB3R repressors control downregulated genes [12] — consistent with the user's finding that cell cycle genes (downregulated by radiation) are SOG1-independent (interaction-UP). Yoshiyama et al. (2017) showed that SOG1 phosphorylation at SQ motifs incrementally tunes the DDR [16], providing a mechanism for the dose-dependent DDR activation seen across the user's datasets.

3.2 Radiation Quality Effects

Dixit et al. (2023) found that simulated GCR activates DNA repair pathways and downregulates glucosinolate biosynthesis in Arabidopsis seedlings [2]. The user's OSD-658 enrichment data directly confirms this: DNA repair is the sole upregulated pathway at both 40 and 80 cGy GCR, and glucosinolate biosynthesis is the top downregulated pathway at 40 cGy. The near-zero gene overlap between GCR and gamma studies in the user's data is a novel finding that extends Dixit et al.'s work by demonstrating the qualitative difference at the gene level, not just the pathway level.

3.3 Cross-Study Confounders

Barker et al. (2023) found that assay type (microarray vs. RNA-seq) is the largest confounding factor in cross-study plant spaceflight comparisons, and that radiation treatment is the most correlated biological factor [4]. The user's data, all generated by RNA-seq through a common DESeq2 pipeline, avoids the assay-type confound. However, developmental stage (6-day seedlings vs. 10-day seedlings vs. 4-week rosettes) and growth conditions (dark vs. light, phytagel vs. soil) remain as confounders that must be accounted for in the dose-response meta-analysis.

3.4 Ethylene and Epigenetic Regulation

Newman et al. (2023) identified ethylene signaling as the key pathway in low-dose gamma response, with dose-dependent epigenetic changes at ERF binding sites [21]. The user's OSD-782 enrichment data shows defense regulation and hypoxia as top upregulated pathways at both 10 and 100 cGy — both are downstream of ethylene signaling. The 148 shared upregulated GO terms between 10 and 100 cGy provide the pathway-level substrate for integrating the OSD-782 epigenetic data (MAPit chromatin accessibility, bisulfite methylation) with the user's DEGs.

4. Targeted Analytical Approaches (Revised with Data)

The six approaches below are revised based on the actual cross-study findings. Each now references specific data points from the 11 comparisons and prioritizes analyses where the data already shows strong signals.

Approach 1: SOG1 Regulatory Network Characterization

Priority: Highest — the 5b data already provides the key result

What the data shows: 46 SOG1-dependent radiation-responsive genes, including 11 known SOG1 direct targets. The interaction-DOWN pathways (DNA repair, DDR, DSB repair, HR, response to ionizing radiation) are the canonical SOG1 regulon. The interaction-UP pathways (cell cycle, mitotic, meiotic) are SOG1-independent, likely controlled by MYB3R repressors [12].

What to do next:

1. Annotate all 46 SOG1-dependent genes: which are known targets [14], which are novel candidates?
2. Search for the SOG1 binding motif CTT(N)7AAG in promoters of the 35 novel candidates.
3. Compare the 46 genes against the 146 Ogita et al. targets to identify novel SOG1-dependent genes not found by ChIP-seq (possibly indirect targets).
4. Test whether the 11 known targets show dose-dependent SOG1 dependency across OSD-782 and OSD-658.
5. Generate a SOG1 regulatory network diagram showing direct vs. indirect targets.

R implementation:

```
# The 46 SOG1-dependent genes are already identified
sog1_dep <- intersect(genes_5a_up, genes_5b_int_down)
# Known targets from Ogita et al. 2018 Table S2
known_targets <- c("AT5G20850", "AT4G21070", "AT2G31320", ...) # 146 genes
novel_candidates <- setdiff(sog1_dep, known_targets) # 35 genes
# Motif search in novel candidates
library(Biostrings)
promoters <- getPromoterSeqs(genes=novel_candidates, upstream=1000)
motif_hits <- matchPWM(PWM("CTTNNNNNNNAAG"), promoters)
```


Expected outputs: SOG1 dependency map (direct vs. indirect targets), motif enrichment in novel candidates, dose-dependent SOG1 dependency analysis, regulatory network figure. **Publication value:** This is the most novel finding — direct experimental validation of SOG1 dependency under gamma radiation, with 35 novel candidate targets. Sufficient for a mechanistic paper in Plant Cell or Plant Journal.

Approach 2: Radiation Quality Comparison (Gamma vs. GCR)

Priority: High — the near-zero gene overlap is a striking, novel finding

What the data shows: Gene overlap between gamma and GCR studies is essentially zero (Jaccard < 0.005). GCR activates a narrow DDR-focused response (2-7 genes, 26-48x FE), while gamma activates broad defense/hypoxia programs (798-1,578 genes). This is not a dose effect — the GCR doses (40-80 cGy) overlap with the OSD-782 gamma doses (10-100 cGy). The difference is radiation quality: mixed high-LET particles (GCR) vs. low-LET gamma photons.

What to do next:

1. Download raw counts for OSD-658 and OSD-782 from GeneLab.
2. Normalize through a common pipeline (STAR -> featureCounts -> DESeq2).
3. At matched doses (40-80 cGy), directly compare gamma vs. GCR DEG lists.
4. Test whether the GCR-exclusive DDR genes (RAD51, BRCA1, PARP2, TSO2, XRI1, RAD17, TK1A) show any response under gamma at matched doses — are they truly GCR-specific?
5. Examine whether the glucosinolate downregulation (GCR40) occurs under gamma at any dose.
6. Model LET-dependent pathway activation using the fold enrichment differences.

Expected outputs: Gamma vs. GCR DEG comparison at matched doses, LET-dependent pathway activation model, radiation-quality-specific gene lists. **Publication value:** The qualitative difference between gamma and GCR plant responses is directly relevant to NASA space radiation risk assessment. This finding alone could anchor a space biology paper in Frontiers in Plant Science or NPJ Microgravity.

Approach 3: Dose-Response Curve Fitting

Priority: Medium — requires raw count integration across studies

What the data shows: The GO enrichment data reveals dose-dependent pathway activation. Hypoxia response strengthens from 10 cGy (FE=3.0) to 100 cGy (FE=4.0) to 100 Gy (FE=3.5). Defense response appears at 10 cGy but becomes the dominant signal at 100 Gy. DNA repair is absent from gamma upregulated sets at low dose but appears in GCR upregulated sets and in the 100 Gy upregulated set. Ribosome biogenesis downregulation strengthens from 100 Gy (FE=2.7) to 10 cGy (FE=5.3).

What to do next:

1. Download raw counts for OSD-782 and OSD-658.
2. Apply ComBat-seq for cross-study batch correction.
3. For key DDR genes (RAD51, BRCA1, PARP1/2, TSO2, XRI1), fit dose-response curves across all available doses (10 cGy, 100 cGy, 40 cGy GCR, 80 cGy GCR, 100 Gy).
4. Test linear, quadratic [28], and threshold models.
5. Compare dose-response shapes between gamma and GCR for genes present in both.

Expected outputs: Dose-response curves for 8-10 key genes, model comparison (linear vs. quadratic vs. threshold), gamma vs. GCR dose-response comparison. **Publication value:** A dose-response meta-analysis spanning 5 orders of magnitude is unprecedented in plant radiation biology and directly addresses the NASA-relevant question of low-dose extrapolation.

Approach 4: Consensus Radiation Signature via WGCNA

Priority: High — defines the core finding alongside SOG1

What the data shows: Only 2 GO terms are upregulated in 4+ of 5 radiation comparisons (programmed cell death, cell death), while 59 GO terms are downregulated in 3+ of 4 comparisons. This asymmetry suggests the downregulated response (translation, cell cycle, plastid) is the more conserved radiation signature, while the upregulated response is dose- and quality-specific. Gene-level overlap between gamma studies is moderate (Jaccard 0.16-0.43), providing enough shared genes for WGCNA module identification.

What to do next:

1. Combine normalized data from all gamma studies (OSD-498/510, OSD-782).
2. Run WGCNA to identify co-expression modules.
3. Correlate modules with radiation vs. control, dose, and developmental stage.
4. Identify the consensus downregulated module (translation/cell cycle/plastid) as the core radiation stress signature.
5. Test this signature against spaceflight datasets from Barker et al. [4] to distinguish radiation-specific from general spaceflight stress.
6. Examine whether GCR data fits into any gamma-derived modules (expected: no, based on near-zero gene overlap).

Expected outputs: Consensus radiation signature gene list, WGCNA module-trait correlation heatmap, radiation vs. spaceflight comparison. **Publication value:** A validated consensus signature, distinguished from general spaceflight stress, is the kind of integrative finding that elevates a paper from single-study report to synthesis.

Approach 5: Glucosinolate-Immunity Trade-off

Priority: Medium — confirmed in GCR data, novel space agriculture angle

What the data shows: Glucosinolate biosynthesis is the top downregulated pathway at GCR 40 cGy (FE=323x, $p=5.2e-05$), confirming Dixit et al. (2023) [2]. Indole glucosinolate metabolic process is upregulated in the 100 Gy gamma set (FE=3.8, $p=1.8e-04$), suggesting the glucosinolate response may be radiation-quality-dependent (down in GCR, up in high-dose gamma). Defense response to bacterium is the top upregulated pathway at 100 Gy (193 genes) and is also upregulated at 10/100 cGy gamma, but is absent from GCR responses.

What to do next:

1. Extract glucosinolate pathway genes (aliphatic: CYP79F2, BCAT4, MAM3, BAT5; indolic: CYP79B2/B3).
2. Track their expression across all 11 comparisons.
3. Test for reciprocal regulation with glutathione metabolism genes.
4. Examine MYB28/MYB29 (glucosinolate activators) expression across doses and radiation types.
5. Correlate glucosinolate downregulation with defense pathway upregulation — is there a trade-off?

Expected outputs: Glucosinolate pathway expression map across studies, glutathione trade-off analysis, radiation-quality-dependent glucosinolate response. **Publication value:** The glucosinolate-immunity trade-off has direct implications for space agriculture (pathogen defense and nutrition in space-grown crops). This angle differentiates the paper from standard DDR-focused studies.

Approach 6: Epigenetic Integration with OSD-782

Priority: Lower — most computationally intensive, highest novelty

What the data shows: OSD-782 enrichment data shows 148 shared upregulated GO terms between 10 and 100 cGy, including defense regulation, hypoxia, and immune signaling — all downstream of ethylene signaling [21]. The 10 and 100 cGy responses diverge in later pathways, consistent with Newman et al.'s finding of dose-dependent epigenetic divergence at 72 hours. The shared pathways provide the gene sets for epigenetic integration.

What to do next:

1. Download OSD-782 MAPit (chromatin accessibility) and bisulfite-seq (DNA methylation) data.
2. Identify DARs and DMRs at 10 and 100 cGy.
3. Map DAR/DMR-associated genes and test overlap with the 148 shared upregulated GO term genes.
4. Perform motif enrichment in DARs near defense/hypoxia genes (expected: ERF, WRKY motifs [21]).
5. Test whether the 46 SOG1-dependent genes show concordant chromatin remodeling.
6. Integrate with the user's high-dose DEGs to test epigenetic concordance across doses.

Expected outputs: Epigenetic-transcriptomic concordance plot, TF motif enrichment in accessible regions, integrative genome browser figures. **Publication value:** Multi-omics integration linking epigenetic regulation to the SOG1 network and dose-dependent pathway activation is at the frontier of plant radiation biology.

5. Metadata Questions Addressed by the Data

The 11 comparisons directly address several metadata questions that were previously open. The table below maps each question to the data evidence and remaining analyses.

#	Metadata question	Data evidence	Next step
Q 1	Is the defense/immune upregulation SOG1-dependent?	PARTIALLY ANSWERED: 5b shows defense genes are NOT in interaction-DOWN (SOG1-independent), while DDR genes ARE SOG1-dependent. SOG1 controls DDR, not defense.	Approach 1
Q 2	Do DDR genes show linear or threshold dose responses?	PARTIALLY ANSWERED: DDR genes appear in GCR up (40-80 cGy) and 100 Gy up, but in 10 cGy DOWN. Non-monotonic dose response suggested.	Approach 3
Q 3	Does GCR produce different responses than gamma?	ANSWERED: Yes. Near-zero gene overlap. GCR = narrow DDR focus; gamma = broad defense/hypoxia. Radiation quality matters more than dose.	Approach 2
Q 4	Is the radiation response conserved across doses?	PARTIALLY ANSWERED: Downregulated (ribosome, cell cycle, plastid) is conserved (3/4 studies). Upregulated is dose-specific (only PCD in 4/5).	Approach 4
Q 5	What is SOG1's role under real gamma (not zeocin)?	ANSWERED: 46 SOG1-dependent genes, 11 known targets confirmed. SOG1 controls DDR/HR under gamma. Cell cycle is SOG1-independent.	Approach 1
Q 6	Is glucosinolate downregulation radiation-quality-specific?	PARTIALLY ANSWERED: Down in GCR40, up in 100 Gy gamma. Suggests radiation-quality-dependent glucosinolate response.	Approach 5
Q 7	Are the 35 novel SOG1-dependent genes direct or indirect targets?	NOT YET ANSWERED: Requires motif search (CTT(N)7AAG) in promoters and comparison with ChIP-seq peaks [14].	Approach 1
Q 8	Does the consensus signature differ from spaceflight stress?	NOT YET ANSWERED: Requires WGCNA + comparison with Barker et al. [4] spaceflight gene lists.	Approach 4
Q 9	Which TF binding sites undergo chromatin remodeling?	NOT YET ANSWERED: Requires OSD-782 MAPit data integration with DEGs.	Approach 6

#	Metadata question	Data evidence	Next step
Q 10	Is the low-dose gamma response (10 cGy) qualitatively different from high-dose (100 Gy)?	PARTIALLY ANSWERED: 148 shared GO terms (10 vs 100 cGy), but 100 Gy adds defense as dominant. Core response conserved, amplitude scales with dose.	Approach 3

Table 4: Metadata questions and their status based on the 11 comparisons. Three questions are already answered by the data; four are partially answered; three require additional analysis.

6. Recommended Analysis Sequence

Based on the data findings, the recommended sequence prioritizes the SOG1 characterization (where the data already shows strong results) and the gamma-vs-GCR comparison (the most novel finding) before moving to integrative analyses.

Order	Approach	Key steps	Compute	Est. time	Depends on
1	Approach 1: SOG1 Network	Annotate 46 genes; motif search in 35 novel candidates; dose-dependent SOG1 dependency	Low (existing data)	2-3 days	None
2	Approach 2: Gamma vs. GCR	Download OSD-658/782 raw counts; matched-dose DEG comparison; LET-dependent modeling	Medium (data download)	3-5 days	None (parallel with 1)
3	Approach 4: Consensus Signature	WGCNA on gamma studies; module-trait correlation; spaceflight comparison	Medium	2-3 days	Approach 2 (for normalized data)
4	Approach 3: Dose-Response	ComBat-seq normalization; curve fitting for key DDR genes across 5 doses	Medium	2-3 days	Approach 2
5	Approach 5: Glucosinolate	Pathway expression tracking; glutathione trade-off; MYB28/29 analysis	Low	1-2 days	Approach 2
6	Approach 6: Epigenetic	Download OSD-782 MAPit/bisulfite; DAR/DMR annotation; motif enrichment	High (epigenomic processing)	5-7 days	Approach 2

Table 5: Recommended execution sequence. Approaches 1 and 2 are independent and can run in parallel. Total estimated time: 2-3 weeks.

Key insight: The data already contains the three strongest findings for the paper: (1) SOG1 dependency validated with 46 genes and 11 known targets, (2) gamma vs. GCR qualitative difference with near-zero gene overlap, and (3) conserved downregulated signature (ribosome/cell cycle/plastid). Approaches 1 and 2 develop these existing findings; Approaches 3-6 extend them. The paper could be structured around these three findings even before completing all six approaches.

7. Publication Framework

7.1 Suggested Figure List

Fig #	Title	Content	Section
1	Cross-study GO enrichment summary	Bar chart of GO terms and genes per comparison (Figure 1 in this document)	Results - Overview
2	Pathway conservation heatmap	20 key pathways x 11 comparisons heatmap (Figure 2 in this document)	Results - Cross-study
3	SOG1 interaction pathways	Bar charts of interaction-UP and interaction-DOWN pathways (Figure 3)	Results - SOG1
4	SOG1-dependent gene evidence	Gene set convergence + known target cross-study presence (Figure 4)	Results - SOG1
5	Dose-response summary	GO counts, Jaccard overlap, pathway conservation by dose (Figure 5)	Results - Dose-response
6	SOG1 regulatory network	Network diagram: direct targets, indirect targets, SOG1-independent genes	Results - Approach 1
7	Gamma vs. GCR comparison	DEG overlap, pathway comparison at matched doses, LET-dependent model	Results - Approach 2
8	Consensus radiation signature	WGCNA modules + radiation vs. spaceflight comparison	Results - Approach 4

7.2 Recommended Paper Structure

Title suggestion: "SOG1-dependent and radiation-quality-specific transcriptomic responses in Arabidopsis: a cross-study analysis from low-dose GCR to acute gamma radiation"

Results structure:

1. Cross-study overview of GO enrichment (Figure 1, Table 1)
2. SOG1 regulatory network: 46 dependent genes, 11 known targets, 35 novel candidates (Figures 3-4, Table 3)
3. Radiation quality: gamma vs. GCR produce fundamentally different responses (Figure 5, Approach 2 results)
4. Conserved downregulated signature: ribosome/cell cycle/plastid across gamma doses (Table 2, Approach 4 results)
5. Dose-response: non-monotonic DDR activation, glucosinolate trade-off (Approaches 3, 5 results)

7.3 Target Journals

Option A (recommended): Plant Cell or Plant Journal — Mechanistic focus on SOG1 network validation under gamma, with radiation quality and dose-response as supporting evidence. The 46 SOG1-dependent genes and 35 novel candidates are sufficient novelty for a mechanistic paper.

Option B: Frontiers in Plant Science or NPJ Microgravity — Space biology focus on the gamma vs. GCR qualitative difference and its implications for space agriculture. The near-zero gene overlap and glucosinolate trade-off are the headline findings.

Option C: Journal of Environmental Radioactivity or Int. J. Radiation Biology — Dose-response focus spanning 10 cGy to 100 Gy, with the non-monotonic DDR activation and conserved downregulated signature as core findings.

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